CSS LAYOUT ARCHITECTURE

Grid vs. Flexbox Guide

1 OVERVIEW

Modern CSS provides two powerful layout systems. Choosing the right tool depends on whether your design problem is one-dimensional or two-dimensional.

2 HOW IT WORKS

FLEXBOX (One-Dimensional)

- Best for a row or column.
- Content-driven (sizes based on content).
- Ideal for navbars and component alignment.

CSS GRID (Two-Dimensional)

- Handles rows and columns.
- Layout-driven (defines grid tracks first).
- Ideal for page layouts, dashboards and complex structures.

3 CORE DIFFERENCE

FLEXBOX	CSS GRID
One-dimensional	Two-dimensional
Row OR column	Rows AND columns
Content-driven	Layout-driven
Best for components	Best for full page layouts

4 EXAMPLE: CHOOSING THE RIGHT LAYOUT

FLEXBOX (1D)

Single row or single column



Row (horizontal)



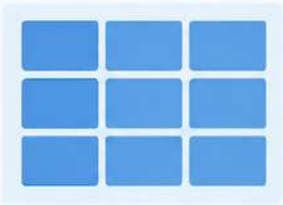
Column (vertical)

FLEXBOX PROPERTIES

```
display: flex;
flex-direction: row | column;
justify-content: space-between;
align-items: center;
flex-wrap: wrap;
```

CSS GRID (2D)

Multiple rows and columns



GRID PROPERTIES

```
display: grid;
grid-template-columns: ...;
grid-template-rows: ...;
gap: 1rem;
grid-template-areas: ...;
```

DECISION CHEAT SHEET

1. Are you aligning items along a single axis (row or column)?
   → USE FLEXBOX
2. Do you need to control row tracks and column tracks simultaneously?
   → USE CSS GRID
3. Can you use both together?
   YES! Grid handles the macro page layout, while Flexbox handles the micro component alignment inside individual grid cells.

5 LIMITATIONS

- ✓ Flexbox is not ideal for complex 2D layouts.
- ✓ Grid can be overkill for simple components.
- ✗ Both require modern browser support.
- ✗ Learning curve for advanced features.

KEY TAKEAWAY

Use Flexbox for one-dimensional layouts, use Grid for two-dimensional layouts, and combine both when needed.

"Master the axis, control the space."